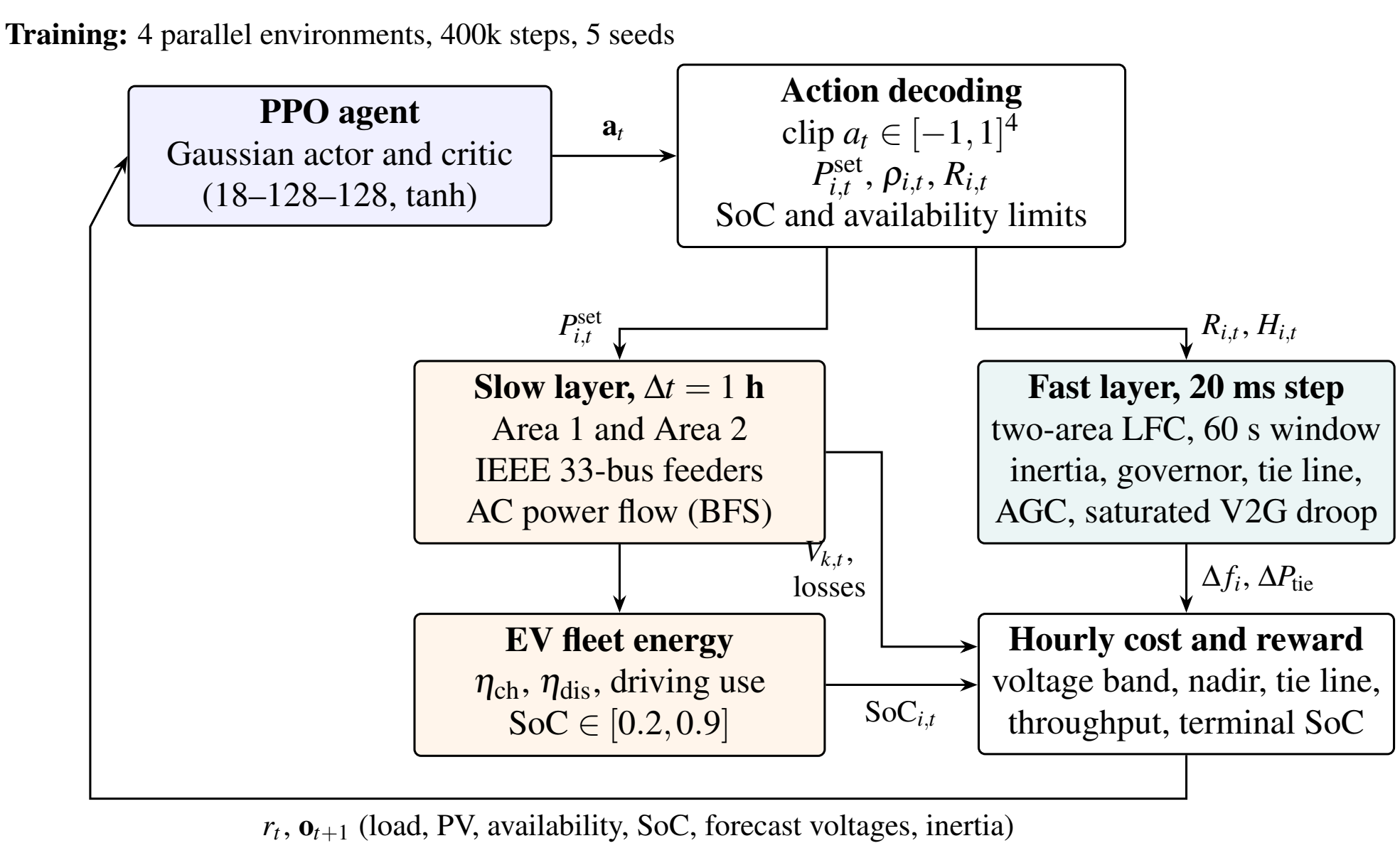

Training: 4 parallel environments, 400k steps, 5 seeds
PPO agent
Gaussian actor and critic
(18–128–128, tanh)
$\mathbf{a}_t$
Action decoding
clip $a_t \in [-1,1]^4$
$P^{\text{set}}_{i,t}$, $\rho_{i,t}$, $R_{i,t}$
SoC and availability limits
$P^{\text{set}}_{i,t}$
$R_{i,t}$, $H_{i,t}$
Slow layer, $\Delta t = 1$ h
Area 1 and Area 2
IEEE 33-bus feeders
AC power flow (BFS)
Fast layer, 20 ms step
two-area LFC, 60 s window
inertia, governor, tie line,
AGC, saturated V2G droop
$V_{k,t}$,
losses
Δf_i, ΔP_{tie}
EV fleet energy
η_{ch}, η_{dis}, driving use
SoC $\in [0.2, 0.9]$
$\text{SoC}_{i,t}$
Hourly cost and reward
voltage band, nadir, tie line,
throughput, terminal SoC
r_t, $\mathbf{o}_{t+1}$ (load, PV, availability, SoC, forecast voltages, inertia)